

CSMFAB0000000000009 Aegis Home_ BFRP Rebar Fabrication Plan

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMFAB0000000000009 Aegis Home_ BFRP Rebar Fabrication Plan V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
 3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete
-

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
BFRP rebar design standard	V1.0: ACI 440.1R-15	V2.0: ACI 440.11-25 (Seismic Type S); design tensile 1000 MPa	Seismic provisions absent in V1.0 — required for CA/PNW deployment; 11-year code update	ACI 440.11-25 January 2025; ACI 318-25

Parameter	V1.0	V2.0	Reason	Primary Source
Composite matrix — recyclability	V1.0: Thermoset epoxy; irreversible crosslink; landfill end-of-life	V2.0: Elium thermoplastic acrylic; thermal solvolysis 350°C; 95% fiber + 100% monomer recovery	Thermoset blocks Phoenix Protocol circular economy mandate	Arkema Elium TDS 2025; Composites Part A 2024
EMI/FSS shielding layer	V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz	V2.0: Ti3C2Tx MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz	2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop	ACS Appl. Mater. Interfaces 17/8 (2025) 12234

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
BFRP rebar design standard	Non-compliant with SDC D/E/F seismic zones; limits geographic deployment	HIGH
Composite matrix — recyclability	No end-of-life recovery path; contradicts Phoenix Protocol; Phoenix Protocol breach	HIGH
EMI/FSS shielding layer	47 dB shielding deficit; AgNW susceptible to oxidation degradation over time	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and

Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000009 Aegis Home_ BFRP Rebar Fabrication Plan V1.0 archived; V2.0 is the active specification as of June 2026.